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Synopsis
Smart Wireless EV Charging System

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering / Master of Technology
In
Electronic & Computer Science
by

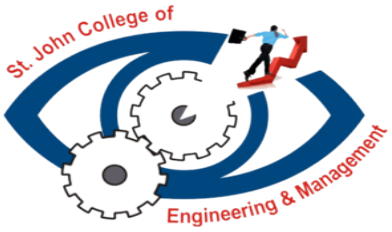
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DEPARTMENT OF ELECTRONIC & COMPUTER SCIENCE
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Certificate of Approval

This is to certify that **Kanak Marde, Prem Meher, Siddhant Meher & Vedant Pagdhare** are bonafide students of B. Tech in **Electronic & Computer Science**-Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Synopsis as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on **"Smart Wireless EV Charging System".**

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Declaration

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2. Prem Meher	----------
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ABSTRACT

Electric vehicles (EVs) are becoming common, but charging them with a cable is not always easy. Cables must be plugged in by hand, connectors wear out, and wet or crowded places can be unsafe. Wireless charging removes the cable by sending power through a magnetic field between two coils.

This project, “Smart Wireless Electric Vehicle Charging System”, builds a small working model of wireless charging and adds smart monitoring to it. Power from the 230 V AC supply is changed to 5 V DC by an SMPS adapter. A transmitter (TX) coil sends this power across an air gap to a receiver (RX) coil. A boost CC/CV converter raises the voltage to about 14.4 V, which charges a 12 V, 7 Ah sealed lead acid (SLA) battery.

An ESP32 microcontroller, an INA226 sensor and a DS18B20 temperature sensor measure voltage, current, power and temperature. The ESP32 estimates battery state of charge (SOC) and the charging status, and sends them live to the Blynk mobile app. The project also studies how efficiency changes with coil distance and coil misalignment. The model is a low-power prototype for learning and testing, and the same idea can be scaled up for real EV chargers.

Keywords: Wireless power transfer, inductive coupling, electric vehicle, ESP32, INA226, Blynk, IoT monitoring, SLA battery.

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Abbreviations and symbols

- **AC:** Alternating Current
- **DC:** Direct Current
- **SMPS:** Switched Mode Power Supply
- **TX / RX:** Transmitter / Receiver
- **WPT:** Wireless Power Transfer
- **IPT:** Inductive Power Transfer
- **EV:** Electric Vehicle
- **SLA:** Sealed Lead Acid (battery)
- **CC / CV:** Constant Current / Constant Voltage
- **SOC:** State of Charge
- **IoT:** Internet of Things
- **I2C:** Inter-Integrated Circuit (communication bus)
- **ESP32:** Wi-Fi enabled microcontroller
- **INA226:** Voltage and current (power) monitor sensor
- **DS18B20:** Digital temperature sensor
- **EMF:** Electromagnetic Field
- **WBS:** Work Breakdown Structure
- **BOM:** Bill of Materials
- **SDG:** Sustainable Development Goal
- **η:** Efficiency
- **V, I, P:** Voltage, Current, Power

CHAPTER 1: INTRODUCTION

1.1Background

The world is moving from petrol and diesel vehicles to electric vehicles to reduce pollution and fuel cost. Global electric car sales were about 14 million in 2023, around 18% of all new cars sold, and rose to about 17 million in 2024 [1]. More EVs mean more charging points are needed, and charging must be easy and safe.

Today most EVs are charged with a plug and cable (wired or conductive charging). The driver has to park, take out the cable, and plug it in. Cables and connectors get damaged by weather, wear and rough use.

Wireless charging, also called Inductive Power Transfer (IPT), works on Faraday's law of electromagnetic induction. An alternating current in the transmitter coil makes a changing magnetic field. This field passes through the air gap and induces a voltage in the receiver coil. The induced AC is then converted to DC to charge the battery [2], [3]. For EVs, the standard SAE J2954 uses a frequency of about 85 kHz [9]; small low-power modules like the one used in this project work at roughly 100 to 200 kHz.

This project makes a small model of such a system for a 12 V battery and adds an IoT monitoring unit (ESP32 + sensors + Blynk app) so that the charging can be seen and studied live.

1.2 Problem Statement

- **Existing problem:** Wired EV charging needs manual plugging, and connectors and cables wear out. It is difficult to automate in parking areas.
- **Industrial issue:** Wireless charging is less efficient than wired charging (roughly 70 to 85% against 90% or more), is sensitive to coil alignment, and works only over a short air gap. Coil losses produce heat.

- **Research gap:** Most studies focus on high-power design. Few give a low-cost, easy-to-repeat model with live monitoring of voltage, current, power, temperature and SOC.
- **Need for improvement:** A simple, affordable system is needed to measure and study how wireless charging performs in real conditions, so that its weak points (distance, misalignment, heat) can be understood.

1.3 Need of the Project

- EV use is growing fast, so easier and safer charging methods are needed.
- Wireless charging removes cables and connectors, so there is less wear and less risk of electric shock in wet places.
- Wireless charging can be made fully automatic, which suits parking lots and fleet depots.
- Its efficiency and heat loss must be measured and understood before it can be used widely.
- Live monitoring through IoT helps users and engineers watch the charging process and receive alerts.

1.4 Objectives

General objective

To design, build and test a small wireless charging system for a 12 V battery with live IoT monitoring, and to study its performance.

Specific objectives

- Design the power circuit: 230 V AC to 5 V DC, wireless TX/RX coils, and boost CC/CV converter to charge a 12 V, 7 Ah SLA battery.
- Develop the monitoring unit using ESP32, INA226 and DS18B20 to measure voltage, current, power and temperature.

- Estimate the battery SOC and show the charging / not charging status on the Blynk mobile app.
- Fabricate and assemble the complete working prototype with a fuse and switch for safety.
- Test the efficiency of the system for different coil distances and different misalignments.
- Observe the temperature rise during charging and set a safety alert limit.

1.5 Scope

What will be covered

- A low-power prototype (5 V, 2 A input) charging a 12 V, 7 Ah SLA battery.
- Live monitoring of V, I, P, temperature, estimated SOC and charging status.
- Study of efficiency against coil distance and coil alignment.
- Basic safety: fuse, switch and temperature alert.

Limitations

- It is a small model. It does not charge a real EV, and the power level is about 10 W at most.
- Charging is slow because of the low power. A 7 Ah battery will take many hours to charge fully.
- SOC is only an estimate based on battery voltage and current. It is not an exact value.
- The air gap is short (a few centimetres) and the coils must be nearly aligned.

Future expansion

- Higher power coils and a custom high-frequency inverter and rectifier.
- Automatic coil alignment, smart slot booking, and solar power input.
- Battery health prediction using charging history data.

1.6 Expected Outcomes

- A working prototype of wireless battery charging with IoT monitoring.
- Microcontroller software (ESP32 program) and a Blynk dashboard.
- Graphs of efficiency against distance and misalignment, and of temperature rise.
- A project report, and a possible paper or poster for a student conference.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Wireless power transfer has been studied for more than a decade, from small mobile chargers to large systems for EVs and even moving vehicles on roads. The papers below were studied to understand the working principle, the design methods, the standards and the safety limits. They also helped to find what is still missing.

2.2 Literature Review Table

Paper	Author	Year	Method	Findings	Research Gap
Wireless Power Transfer via Strongly Coupled Magnetic Resonances	A. Kurs et al.	2007	Two resonant coils tested experimentally	Powered a 60 W bulb over about 2 m with around 40% efficiency using magnetic resonance	Short of EV battery charging; no live monitoring
Inductive Power Transfer	G. A. Covic, J. T. Boys	2013	Review of IPT theory, compensation and pad design	Explains how IPT works and how coils are tuned for good power transfer	Focus on high-power design; no low-cost monitoring model
Modern Trends in Inductive Power Transfer for Transportation Applications	G. A. Covic, J. T. Boys	2013	Review of IPT for vehicles	Covers stationary and moving vehicle charging and coil pad types	Little about small test models and data logging

Paper	Author	Year	Method	Findings	Research Gap
Wireless Power Transfer for Electric Vehicle Applications	S. Li, C. C. Mi	2015	Review of WPT types for EVs	Compares inductive and resonant methods; lists efficiency, alignment and cost issues	Does not give a simple method to study alignment and heat live
Modern Advances in WPT Systems for Roadway Powered EVs	C. C. Mi et al.	2016	Review of roadway (dynamic) charging	Shows progress in charging EVs while driving; names cost and EMF as challenges	Large systems only; not suited for student-level study
A Critical Review of Recent Progress in Mid-Range WPT	S. Y. R. Hui et al.	2014	Review of mid-range WPT methods	Shows that efficiency falls as the distance between coils grows	No practical IoT-based measurement of this effect
Design and Implementation of Shaped Magnetic-Resonance-Based WPT for Roadway-Powered Moving EVs	J. Shin et al.	2014	Built and tested an on-road system	Showed that a moving EV can be charged wirelessly across an air gap	High cost; no open low-cost monitoring design
Advances in WPT Systems for Roadway-Powered EVs	S. Y. Choi et al.	2015	Review of roadway-powered systems	Discusses track and pickup design, EMF and cost	No data-driven study on small prototypes

Paper	Author	Year	Method	Findings	Research Gap
SAE J2954: WPT for Light-Duty Plug-in/Electric Vehicles and Alignment Methodology	SAE International	2020	Industry standard	Sets power classes, frequency (about 85 kHz) and alignment methods	Standard only; gives no monitoring tool
Guidelines for Limiting Exposure to Time-Varying Electric and Magnetic Fields (1 Hz to 100 kHz)	ICNIRP	2010	Safety guideline	Gives safe exposure limits for people near magnetic fields	Needs to be checked for each design

2.3 Research Gap

What previous researchers have done

- Proved that power can be sent wirelessly using magnetic coupling and resonance.
- Built high-power systems for parked and moving EVs and made standards for them.
- Studied efficiency, coil shapes and safety limits.

What remains unsolved

- Efficiency drops with distance and misalignment, and there are few simple, low-cost setups to measure this live.
- Coil heating is not always monitored in real time.
- Many works do not combine power transfer with IoT-based monitoring and SOC display.

How this project addresses the gap

This project builds a small, low-cost wireless charging model and adds an ESP32-based monitoring unit. It measures voltage, current, power and temperature, estimates SOC, and shows the data live on a phone. It then uses this data to study the effect of distance and misalignment on efficiency.

2.4 Novelty / Innovation

- **Innovative idea:** Wireless charging of a lead acid battery with complete live IoT monitoring on a mobile app.
- **Unique features:** Live V, I, P, temperature, estimated SOC and charging status; temperature alert; study of distance and alignment.
- **Improvement over existing systems:** Low cost and easy to repeat in a college lab, and it gives measurable data that big commercial systems do not share.

CHAPTER 3: PROPOSED METHODOLOGY

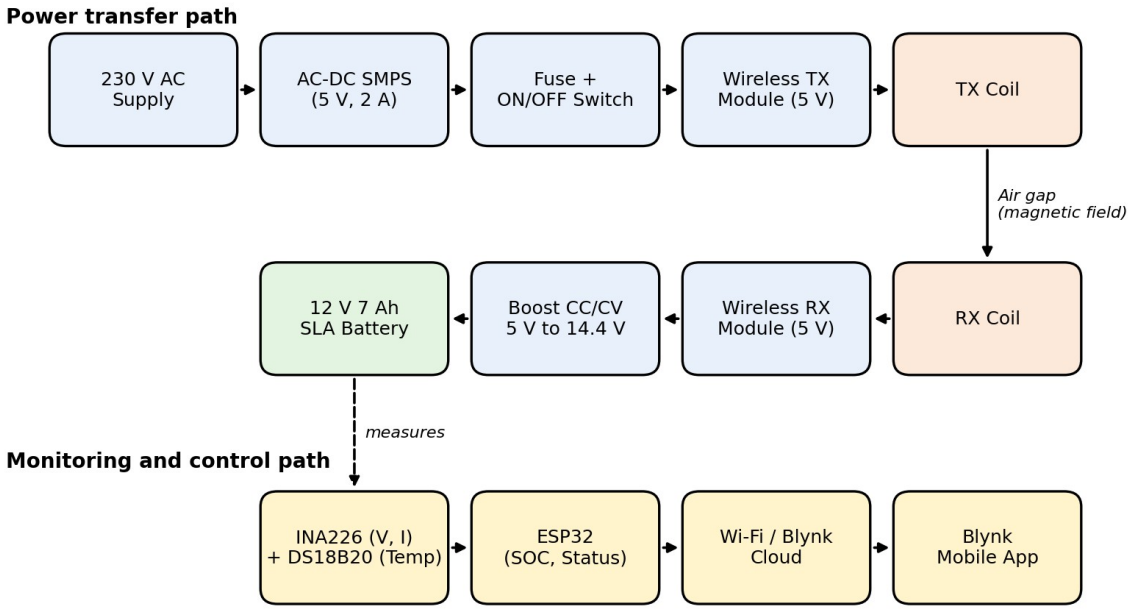
3.1 Proposed System

The proposed system has two paths. The power transfer path takes energy from the 230 V AC supply to the 12 V battery through the wireless coils. The monitoring path uses the ESP32 with the INA226 and DS18B20 sensors to measure the charging and send the data to the Blynk app over Wi-Fi.

3.2 Working Principle

1. The SMPS adapter changes 230 V AC to 5 V DC (2 A).
2. The 5 V DC passes through a fuse and an ON/OFF switch to the wireless TX module.
3. The TX module makes high-frequency AC, and the TX coil produces a changing magnetic field.
4. The magnetic field crosses the air gap and induces a voltage in the RX coil.
5. The RX module changes this AC to 5 V DC.
6. The boost CC/CV converter raises 5 V to about 14.4 V. Constant current (CC) protects the battery at the start, and constant voltage (CV) stops over-charging near the end.
7. The 12 V, 7 Ah SLA battery is charged by this output.
8. The INA226 measures battery voltage and current, and the DS18B20 measures temperature. The ESP32 calculates power and SOC and sends the data to the Blynk app.

3.3 System Architecture / Block Diagram



3.4 DFD

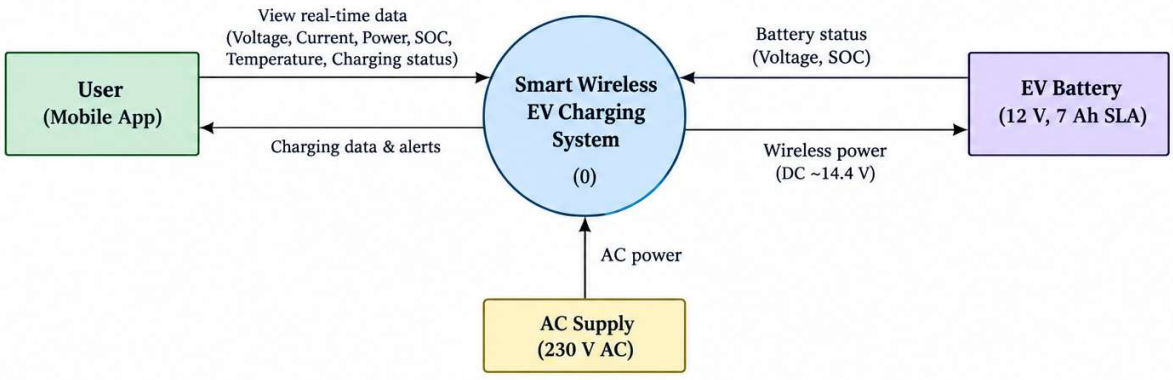
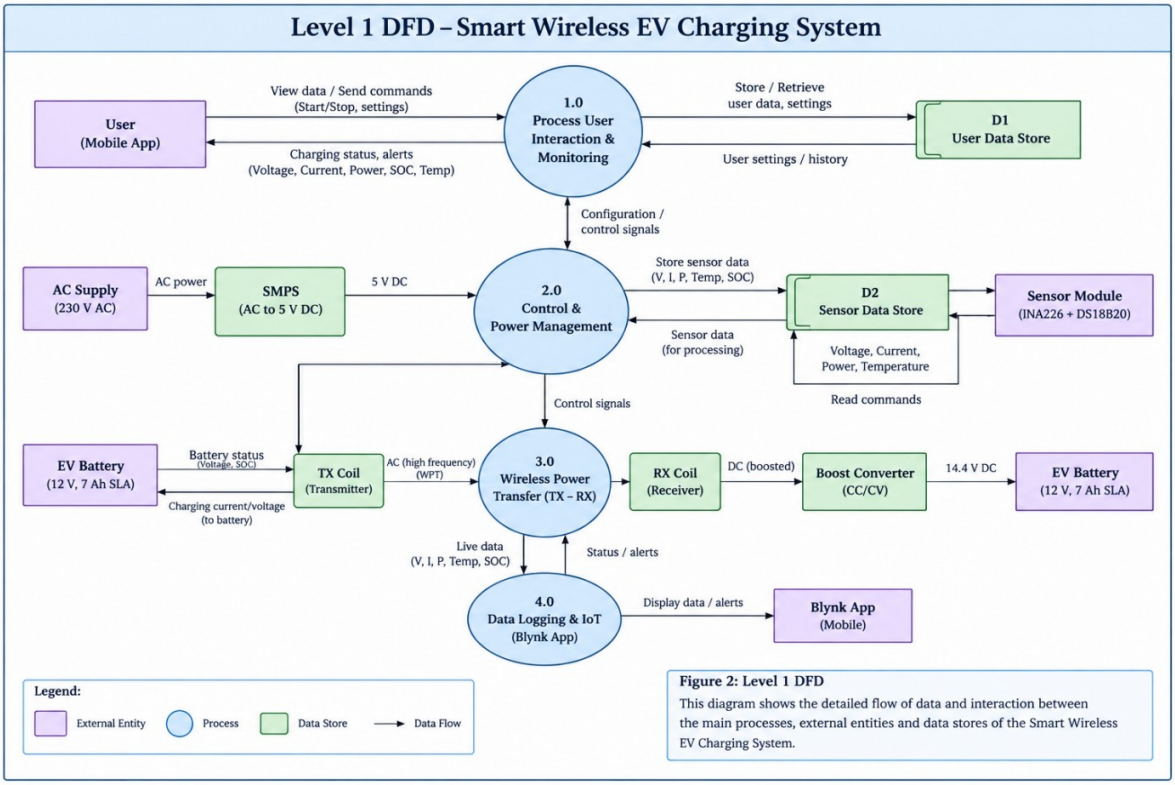
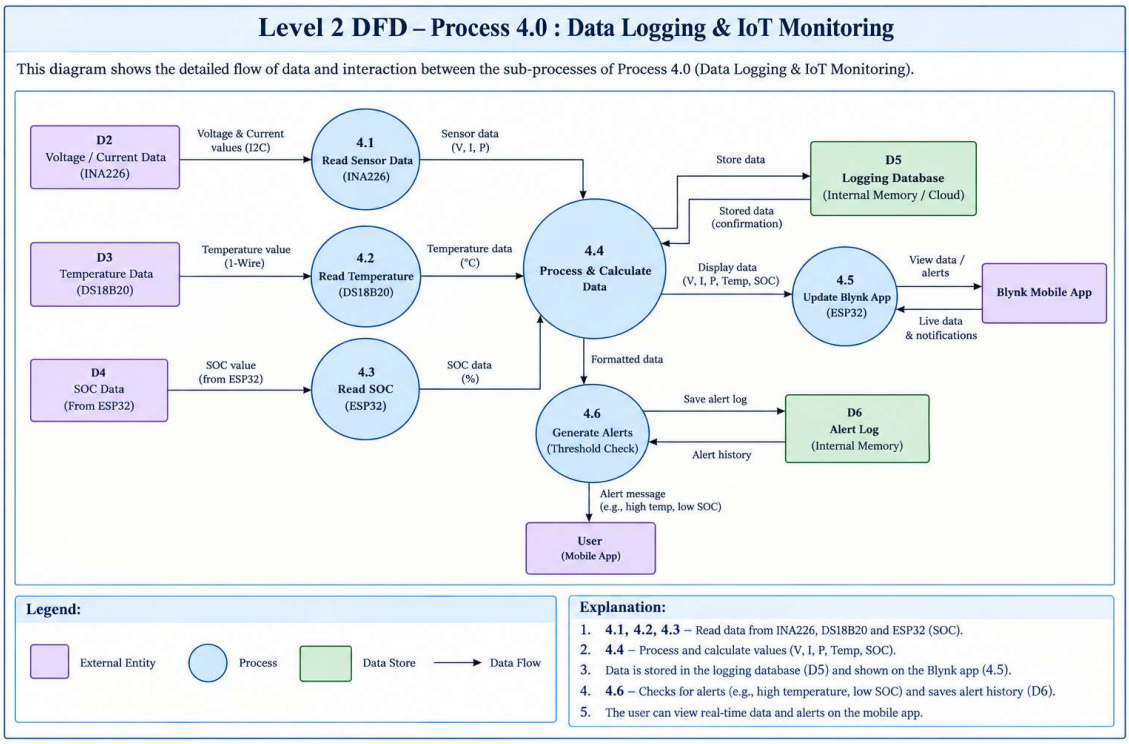


Figure 1: Level 0 – Context Diagram

Level 0 DFD

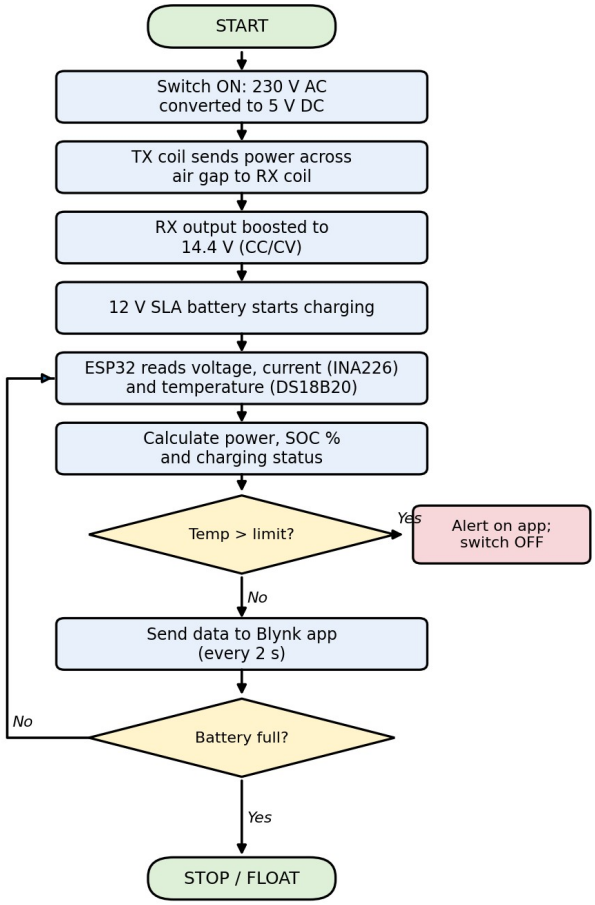


Level 1 DFD



Level 0 DFD

3.5 Flow Chart



3.6 Methodology

Design procedure

- Select the battery (12 V, 7 Ah SLA) and find its charging voltage (about 14.4 V).
- Check the power available: 5 V × 2 A = 10 W at most, so the charging current will be small (about 0.3 to 0.5 A, estimated after losses).
- Select the wireless TX/RX module, the boost CC/CV converter, the fuse (2 to 2.5 A) and the switch.
- Select the sensors: the INA226 measures bus voltage up to 36 V and current through a 0.1 Ω shunt (about 0.8 A maximum, which is enough here); the DS18B20 gives digital temperature.

Experimental procedure

- Assemble the circuit on a bench and check every connection with a multimeter before switching ON.
- Set the boost converter output to 14.4 V (CV) and set a safe current limit (CC) before connecting the battery.
- Measure input power and battery power at different coil distances (for example 0, 5, 10, 15 mm) and different sideways shifts.
- Record temperature of the coil area and the converter during charging.
- Calculate efficiency using $\eta = (P_{out} \div P_{in}) \times 100\%$.

Algorithm (ESP32 programming steps)

1. Start the ESP32, connect to Wi-Fi and to the Blynk server.
2. Initialise the INA226 (I2C) and the DS18B20 (one-wire).
3. Every 2 seconds: read voltage V, current I and temperature T.
4. Calculate power $P = V \times I$.
5. Estimate SOC from a battery voltage table (for a rested 12 V SLA battery, about 12.7 V is near 100%, 12.5 V is about 75%, 12.2 V is about 50%, 12.0 V is about 25% and 11.8 V is near 0%). While charging, the voltage reads higher, so the value is corrected or taken as approximate.
6. Set the status to “Charging” if the current is above a small limit (for example 0.05 A), else “Not charging”.
7. If the temperature goes above the set limit (for example 50 °C), send an alert to the app.
8. Send V, I, P, T, SOC and status to the Blynk app, and repeat.

Assembly (manufacturing) process

- Fix the TX coil on a base and the RX coil on a movable holder so that distance and shift can be changed.
- Wire the power path using 22 to 24 AWG wire and screw terminals; use proper battery wires for the battery side.
- Mount the ESP32 and sensors on a small board and connect the INA226 in series with the battery charging line.

Place everything in a neat insulated box with the switch and fuse easy to Reach

3.7 Components / Materials

Component	Specification	Quantity
AC-DC SMPS adapter	Input 230 V AC, output 5 V DC, 2 A (3 A preferred)	1
Wireless TX-RX module	5 V DC, 2 A rated, with TX coil and RX coil	1 set
Fuse with inline holder	2 A or 2.5 A DC	1
ON/OFF switch	SPST, 5 V DC, 3 A or more	1
Boost CC/CV module	Input about 5 V, adjustable output up to 14.4 V, CC and CV adjustment	1
SLA battery	12 V, 7 Ah, rechargeable SLA / AGM / VRLA	1
ESP32 development board	Wi-Fi and Bluetooth microcontroller	1
INA226 module	I2C voltage / current / power monitor, 0.1 Ω shunt	1
DS18B20 sensor	Digital temperature sensor (waterproof probe) with 4.7 kΩ pull-up resistor	1
Battery wires and connectors	Insulated wires suitable for 12 V battery	1 set
Connecting wires	22 to 24 AWG insulated hook-up wire	As needed
Terminal connectors	2-pin screw terminals / DC connectors	As needed
Digital multimeter	DC voltage and current measurement	1

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

- **1. Study and planning:** problem finalisation, literature survey, block diagram.
- **2. Hardware:** component selection and purchase, power path assembly, coil holder and enclosure.
- **3. Monitoring unit:** sensor connection, ESP32 program, sensor calibration, Blynk dashboard.
- **4. Testing:** charging test, efficiency vs distance, efficiency vs misalignment, temperature test, safety check.
- **5. Documentation:** report, presentation and demonstration.

4.2 Gantt Chart

Semester planning in blocks of two weeks (■ means the activity is active).

Activity	B1	B2	B3	B4	B5	B6	B7	B8
Literature survey and problem finalisation	■	■						
Component selection and purchase		■	■					
Circuit design and block diagram		■	■					
Hardware assembly (power path)			■	■	■			
ESP32 program and sensor calibration				■	■	■		
Blynk dashboard setup					■	■		
Testing: efficiency, distance, alignment, heat						■	■	■
Report writing and presentation							■	■

4.3 Roles and Responsibilities

Student	Responsibility
Siddhant Meher	ESP32 programming and sensor interfacing (INA226, DS18B20)
Prem Meher	Power circuit assembly: SMPS, wireless module, boost converter, battery connections
Vedant Pagdhare	Blynk dashboard, data logging, testing of efficiency, distance and alignment
Kanak Marde	Literature survey, documentation, coil holder and enclosure

4.4 Risk Assessment

Risk	Impact	Mitigation
Battery over-charging	Battery damage, heat, gas release	Set boost output to 14.4 V (CV) and a safe CC limit; ESP32 alert; check with multimeter first
Short circuit or wrong wiring	Burnt module or battery	Fuse in the supply line; double-check polarity; use insulated terminals
High voltage 230 V AC contact	Electric shock	Use a ready enclosed adapter; keep the AC side covered and away from the test area
Overheating of coils or converter	Loss of efficiency, damage	DS18B20 temperature monitoring and alert; ventilation; switch OFF at the limit
Coil misalignment or large gap	Very low power transfer	Fixed marks and holder for the RX coil; study the effect on purpose
Wi-Fi or Blynk connection failure	No live data on the app	Show data on serial monitor as a backup; auto-reconnect in the program
Component delay or failure	Project delay	Order early; keep spare sensors and wires

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The project uses ready, widely available modules (wireless TX/RX module, boost converter, ESP32, INA226, DS18B20). The ESP32 supports Wi-Fi, I2C and one-wire, and the Blynk app supports ESP32 directly. The team can wire, program and test it using lab tools such as a multimeter. So the project is technically feasible.

5.2 Economic Feasibility

The total cost is small (about ₹3,700, see the budget table) because low-power standard modules are used. The cost is within the range of a student project, and the parts can be reused for later work.

5.3 Operational Feasibility

The system is easy to use. The user only switches it ON and places the RX coil near the TX coil. The status is seen on the phone. It needs no special training, and the fuse, switch and temperature alert keep the operation safe.

5.4 Budget Estimation

Table 6: Budget estimation (approximate, to be updated with actual bills)

Item	Quantity	Cost (₹, approx.)
AC-DC SMPS adapter 5 V 2 A	1	250
Wireless TX-RX module with coils	1 set	700
Fuse and holder	1	40
ON/OFF switch	1	40
Boost CC/CV module	1	350
12 V 7 Ah SLA battery	1	850
ESP32 development board	1	450

Item	Quantity	Cost (₹, approx.)
INA226 module	1	250
DS18B20 sensor with resistor	1	150
Wires, battery connectors, screw terminals	As needed	300
Enclosure, coil holder and miscellaneous	1	300
Total		3,680

5.5 Bill of Materials (BOM)

S. No.	Item	Specification	Qty
1	AC-DC SMPS adapter	230 V AC to 5 V DC, 2 A	1
2	Wireless TX-RX module	5 V, 2 A, TX coil + RX coil	1 set
3	Fuse + inline holder	2 A / 2.5 A DC	2
4	SPST switch	5 V DC, 3 A or more	1
5	Boost CC/CV module	5 V in, 14.4 V out, CC and CV adjustable	1
6	SLA battery	12 V, 7 Ah	2
7	ESP32 board	Wi-Fi microcontroller	1
8	INA226 module	I2C, 0.1 Ω shunt	1
9	DS18B20 + 4.7 kΩ resistor	Digital temperature sensor	1
10	Wires and connectors	22 to 24 AWG; screw terminals; battery connectors	As needed
11	Digital multimeter	DC V / I	1

CHAPTER 6: EXPECTED RESULTS

- **Prototype:** A working wireless charging model that charges a 12 V, 7 Ah SLA battery without any cable between the supply side and the battery side.
- **Charging performance:** A small charging current (about 0.3 to 0.5 A is estimated from the 10 W input), so charging will be slow but steady, following the CC then CV pattern.
- **Efficiency:** The overall efficiency from adapter to battery is expected to be roughly 50 to 60% (a target to be confirmed by testing). It is expected to be highest when the coils are close and aligned.
- **Distance and alignment:** Efficiency and power are expected to fall as the coil gap and the sideways shift increase. Graphs will show this clearly.
- **Thermal behaviour:** A small temperature rise is expected in the coil area and converter, and the alert will work if the set limit is crossed.
- **Monitoring:** Live voltage, current, power, temperature, estimated SOC and charging status on the Blynk app with about 2 second updates.
- **Simulation / experimental results:** Experimental graphs of efficiency vs distance, efficiency vs misalignment, battery voltage vs time, and temperature vs time.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

Wireless charging is useful for parking lots, bus and delivery fleets, warehouses and robots, where cables are a problem. The project's monitoring ideas (live data, SOC, heat alert) can be used in industrial chargers for safety and maintenance.

7.2 Societal Impact

Cable-free charging is easier and safer for all users, including elderly and disabled people, and in rain or dirty places. It can make EVs more attractive, which helps people shift to cleaner transport.

7.3 Environmental Impact

More EVs mean less air pollution and fewer greenhouse gas emissions. Wireless charging also avoids frequent replacement of worn cables and connectors. Measuring and reducing losses saves energy. The lead acid battery must be disposed of or recycled properly at the end of its life.

7.4 SDG Mapping

☑ SDG 7: Affordable and Clean Energy

☑ SDG 9: Industry, Innovation and Infrastructure

☑ SDG 11: Sustainable Cities and Communities

☑ SDG 12: Responsible Consumption and Production

☑ SDG 13: Climate Action

7.5 Future Scope

- Scale up to higher power (for example an e-bike, e-rickshaw or car) with a custom high-frequency inverter and rectifier.
- Add automatic coil alignment and foreign object detection.
- Add solar power input and smart slot booking for charging stations.
- Use charging history to predict battery health with machine learning.
- Replace the voltage-based SOC with a better SOC method and use a lithium battery.

CHAPTER 8: CONCLUSION

Problem: Wired EV charging needs cables and manual plugging, and wireless charging has issues of efficiency, alignment and heat that are not easy to observe in real time.

Proposed solution: A small wireless charging system that charges a 12 V, 7 Ah SLA battery through TX/RX coils and a boost CC/CV converter, with an ESP32, INA226 and DS18B20 monitoring unit that shows voltage, current, power, temperature, SOC and status on the Blynk app.

Expected contribution: A low-cost, repeatable model that gives measured data on efficiency, distance, alignment and heating. It builds a base for future higher-power wireless EV charging work and gives the team practical skills in power electronics, embedded systems and IoT.

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